

Assignment-2.1

“Overview of Peripherals theory”

TASK-1 Create a table listing 3 types of printers, 2 types of scanners, and 2 types of webcams, and for each, write one real-life use case (for example: 'Inkjet printer – printing college assignments').

Device	Types	Real life uses
Printer	Inkjet printer	Printing college assignments and color photos
Printer	Laser printer	Printing many black-and-white documents in an office
Printer	3d printer	Physical 3D plastic models and engineering prototypes.
Scanners	Flatbed scanner	Scanning certificates and documents
Scanners	Barcode/ hand-held scanner	Scanning product barcodes rapidly at retail checkout counters.
Webcam	Integrated / Desktop USB Webcam	Attending online lectures, Zoom calls, and video interviews.
Webcam	Conference / PTZ Webcam	Capturing wide-angle video for multi-person university meeting rooms.

TASK-2 Draw a labeled diagram showing how a typical home printer connects to a laptop or PC, including the connection interface used (USB, Wi-Fi, or Bluetooth).



Diagram Components:

1.Host Device (Laptop/PC): Initiates print jobs via the operating system print spooler.

2.Connection Interface Options:

- **Wired:** USB Type-A to Type-B Cable (direct point-to-point hardware connection).
- **Wireless:** Wi-Fi 802.11 b/g/n (local router network) or Bluetooth.

3.Peripheral (Home Printer): Processes incoming print data and outputs physical paper documents.

TASK-3 Draw a labeled diagram showing how a typical home printer connects to a laptop or PC, including the connection interface used (USB, Wi-Fi, or Bluetooth).

USB (Universal Serial Bus):- It establishes a direct wired connection via a dedicated cable between a computer and a peripheral. Its main advantage is fast data transfer speed and high reliability, ensuring lag-free operation without wireless interference. However, its disadvantage is physical restriction; cable length limits device placement and creates desktop clutter while preventing easy multi-device sharing.

Wi-Fi :- It connects devices wirelessly using local area network radio signals. Its major advantage is maximum convenience, enabling multiple computers and smartphones across a building to share a single device without cables. Its key disadvantage is potential network instability, where distance, walls, or router congestion can drop connections or slow down transmission speeds.

Bluetooth:- It provides direct, short-range wireless communication between devices. The primary advantage is quick, direct pairing that operates without needing a router or internet connection. Its main disadvantage is performance limits, offering significantly lower speeds and a limited operational range (under 10 meters) compared to USB or Wi-Fi.

TASK-4 Pick any one device you use at home or college (printer, scanner, or webcam). Write step-by-step instructions for how you would connect it to your computer, including what cable or wireless method you would use and any setup steps required.

Device: HP DeskJet Inkjet Printer

Connection Method: Wi-Fi (with USB fallback)

1. Unpack and Plug In: First, I unbox the printer, peel off all the protective plastic tape, and load some paper into the input tray. Then, I plug the power cable into the wall and turn the printer on.
2. Set Up the Connection: Since I want to print wirelessly, I press and hold the Wi-Fi button on the printer until the network light starts blinking, showing it's ready to pair. (If wireless isn't working, I just run a standard USB cable from the back of the printer directly into my laptop's USB port.)
3. Download the Driver Software: Next, I open a web browser on my laptop, head over to 123.hp.com, and download the official setup app for my printer.
4. Pair with the Computer: I open the installer and follow the prompts on my screen. I select my home Wi-Fi network from the list, type in the Wi-Fi password, and let the laptop find and link up with the printer.
5. Run a Test Print: To make sure everything is working, I go to my laptop's Settings > Devices > Printers & Scanners, select the new printer, and hit Print Test Page to confirm the page comes out cleanly.